FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

Project Overview

The dashboard analyzes:

1. Sales (Total Sales, Quantity)
2. Customer data (Customer Id, Age, Gender)
3. Mall data (Shopping Mall, Area)
4. Time (Invoice Date, Month, Year)

Dashboard Sections

1. Sales Overview Dashboard

1. Total Sales
2. Total Orders
3. Total Quantity
4. Monthly Sales Trend
5. Sales by Mall
6. Sales by Gender
7. Yearly Sales

2. Customer Analysis Dashboard

1. Total Customers
2. Avg Customer Spend
3. Top Customer Sales
4. Age Analysis (Bins)
5. Payment Method Analysis
6. Gender Contribution (Pie)

3. Product / Mall Performance Dashboard

1. Total Quantity Sold
2. Total Malls
3. Avg Sales per Mall
4. Mall Performance
5. Sales vs Quantity
6. Sales Heatmap
7. Area vs Sales

Data Requirements

Section	Fields	Tables

Sales	Total Sales, Quantity, Invoice No	Fact_Sales
Customer	Customer Id, Age, Gender, Payment Method	Dim_Customer
Mall	Shopping Mall, Area (Sqm)	Dim_Mall
Time	Invoice Date, Month Name, Year	Dim_Date
Product	Product Key	Dim_Product

Filters

1. Year
2. Shopping Mall
3. Gender
4. Month Name

All filters support cross-filtering.

Visuals

Dashboard	Visuals
Sales Overview	KPI Cards, Line Chart, Bar Charts
Customer Analysis	Bar Chart, Pie Chart, Histogram
Mall Performance	Scatter Plot, Heatmap, Bar Chart

Interactivity

1. Cross filtering between visuals
2. Drill-down: Year → Month
3. Hover tooltips: Sales, Quantity
4. Dashboard navigation

Calculations

Name	Formula
Total Sales	SUM([Total Sales])
Total Orders	COUNTD([Invoice No])
Total Customers	COUNTD([Customer Id])
Avg Customer Spend	SUM([Total Sales]) / COUNTD([Customer Id])
Avg Order Value	SUM([Total Sales]) / COUNTD([Invoice No])
Avg Sales per Mall	SUM([Total Sales]) / COUNTD([Shopping Mall])

Data Model

1. Star Schema
2. Fact Table: Fact_Sales

3. Dimension Tables: Dim_Customer, Dim_Date, Dim_Mall, Dim_Product
4. One-to-many relationships
5. Keys: Customer Key, Date Key, Mall Key, Product Key

Performance Optimization

1. Use Tableau Extract
2. Optimized joins
3. Minimal calculated fields
4. Efficient filters